representation of a class token [CLS] to classify the image, while auto-regressive based language model (such as GPT) uses the representation of the previous tokens to predict the next word. Consequently, we will focus on the feature (or representation) of this particular token in $\boldsymbol{H}^{(\ell)}$, denoted by $\boldsymbol{h}^{(\ell)} \in \mathbb{R}^d$ at the ℓ-th layer. A linear classifier $g(\cdot)$ is applied to the last layer feature $\boldsymbol{h}^{(L)}$ to make predications as[2] $g(\boldsymbol{h}^{(L)}) = \arg\max_j [\mathrm{SoftMax}(\boldsymbol{W}\boldsymbol{h}^{(L)})]_j$, where $[\cdot]_j$ denotes the j-th entry and $\boldsymbol{W} \in \mathbb{R}^{K\times d}$ maps the d dimensional features to K dimensional logits. We may also directly apply the last layer classifier $g(\cdot)$ to the hidden layer features $\boldsymbol{h}^{(\ell)}$ to make predictions via $g(\boldsymbol{h}^{(\ell)}) = \arg\max_j [\mathrm{SoftMax}(\boldsymbol{W}\boldsymbol{h}^{(\ell)})]_j$. Given data samples $\mathcal{S} := \{\boldsymbol{x}_{k,i}\}$, where $\boldsymbol{x}_{k,i}$ represents the i-th sample of class k with $i \in [n] := \{1, \ldots, n\}$ and $k \in [K]$, we define layer-wise accuracy as

$$\mathrm{Acc}_{\mathcal{S}}^{(\ell)} := \frac{1}{Kn} \sum_{k=1}^{K} \sum_{i=1}^{n} \mathbb{1}[g(\boldsymbol{h}_{k,i}^{(\ell)}) = k]. \tag{2}$$

Existing work on measuring representational similarity Similarity analysis is widely applied in the literature, including research on learning dynamics [35, 36], effects of width and depth [37], differences between supervised and unsupervised models [38], robustness [39, 40], evaluating knowledge distillation [41], language representation [42, 43], and generalizability [44–46]. To enable measuring the similarity of features from from different architectures or layers that have different dimension, most existing methods for analyzing the similarity between representations of high dimensions, such as those based on Canonical Correlation Analysis (CCA) and widely used Centered Kernel Alignment (CKA) [19] , rely on statistical properties of the representations for a set of data points. For instance, given input feature sequence $\boldsymbol{Z}^{\ell} = \begin{bmatrix} \boldsymbol{h}_{1,1}^{(\ell)} & \cdots & \boldsymbol{h}_{K,n}^{(\ell)} \end{bmatrix} \in \mathbb{R}^{d\times N}$ denoting the features of $N = Kn$ training samples, the widely-used CKA with a linear kernel quantifies similarities between features $\boldsymbol{Z}^{\ell}$ and $\boldsymbol{Z}^{\ell'}$ as

$$\mathrm{CKA} = \mathrm{trace}((\boldsymbol{Z}^{\ell'})^\top \boldsymbol{Z}^{\ell'} \cdot (\boldsymbol{Z}^{\ell})^\top \boldsymbol{Z}^{\ell})/(\left\|\boldsymbol{Z}^{\ell}(\boldsymbol{Z}^{\ell})^\top\right\|_F \left\|\boldsymbol{Z}^{\ell'}(\boldsymbol{Z}^{\ell'})^\top\right\|_F). \tag{3}$$

CKA relies on the similarity of inter-example structures since the gram matrix $(\boldsymbol{Z}^{\ell})^\top \boldsymbol{Z}^{\ell} \in \mathbb{R}^{N\times N}$ captures the pair-wise similarity of different samples, focusing on the consistency of relative positions among features. Consequently, the CKA is invariant to orthogonal transformations and isotropic scaling, and can be applied for the case where $\boldsymbol{h}^{(\ell)}$ and $\boldsymbol{h}^{(\ell')}$ have different dimensions.

2.1 Sample-wise layer-wise representational similarity in transformers

In this work, we specifically focus on transformer architectures that obey the following particular properties of features across layers: (i) the features have the same dimension across layers since transformers are generally constructed by stacking identical blocks; (ii) the features may have no or less rotation ambiguity due to the residual connection (1). Based on these observations, for each sample $\boldsymbol{x}_{k,i}$ we propose to simply measure the cosine similarity of the corresponding feature vectors $\boldsymbol{h}^{(\ell)}$ and $\boldsymbol{h}^{(\ell')}$ at layers ℓ and ℓ' as[3]

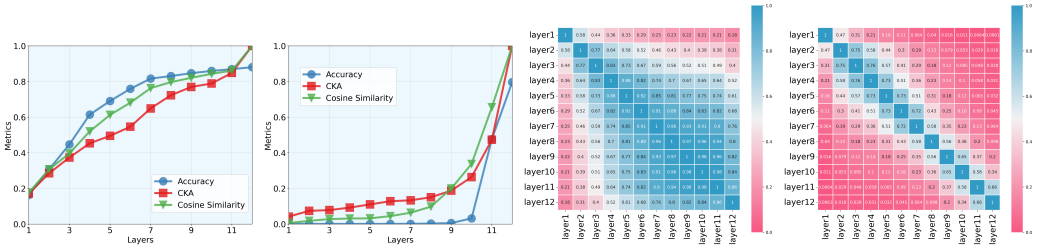
$$\mathrm{COS}(\boldsymbol{h}_{k,i}^{(\ell)}, \boldsymbol{h}_{k,i}^{(\ell')}) = \langle \boldsymbol{h}_{k,i}^{(\ell)}, \boldsymbol{h}_{k,i}^{(\ell')} \rangle / \|\boldsymbol{h}_{k,i}^{(\ell)}\|_2 \|\boldsymbol{h}_{k,i}^{(\ell')}\|_2.$$

The above COsine Similarity (COS) measures the angle between feature vectors, providing a clear geometric interpretation of feature alignment and similarity at the layer level. Unlike CKA (3), COS is not invariant to all transformations except for isotropic scaling. Furthermore, COS is computed for each individual sample and does not rely on inter-example structures. In the experiments, we compute the average COS over all the training samples.

To verify whether the proposed sample-wise COS is a good indicator of similarity structure within transformers, we train the DeiT-S model [24] (a data-efficient vision transform) from scratch on both the CIFAR-10 and ImageNet-1K datasets. The feature dimension is set to 384 for both tasks across all layers. In Figure 2(a, b), we compute CKA and average cosine similarity between the features in each layer and the last layer. Additionally, we plot average COS between all pairs of layers and display the results as a heatmap in Figure 2(c, d). Based on these results, we make several observations.

[2] There is also a bias term $\boldsymbol{b}$ in classification layer, but we omit it for simplicity of presentation.

[3] The features in each layer are centered by reducing the global mean of all the samples.



(a) CIFAR10 (b) ImageNet1K (c) CIFAR10 (d) ImageNet1K

Figure 2: Illustration of a DeiT-S model trained with standard training on CIFAR-10 and ImageNet in terms of (a-b) similarity of features from shallow layers and the last-hidden layer measured by CKA and COS, as well as layerwise validation accuracy, and (c-d) cosine similarities between all pairs of layers. For both datasets, cosine similarity can reflect the trend of layerwise accuracy.

Observation 1: Simple-wise COS is sufficient and reflects CKA to measure layer-wise representation similarity. We observe from Figure 2(a, b) that COS aligns with CKA and is sufficient to reflect the layer-wise representation similarity in transformer models. In other words, since CKA is invariant to orthogonal transformations while COS is not, there is little or no rotation difference among the features from different layers. This is attributed to the residual connections in (1), which create smoother transitions between layers and potentially lead to more stable feature representations throughout the network. In the Appendix B.2, we design additional experiments on multi-layer perceptrons (MLPs) with and without skip connections to verify the effect of skip connections in eliminating rotation ambiguity.

Observation 2: Progressively increasing layer-wise representation similarity. For models trained on small datasets, we observe a ridge-to-plateau pattern in the heatmap plot in Figure 2(c): the initial layers undergo significant transformations, suggesting that lower layers rapidly refine the features to extract the most relevant information for classification tasks; in contrast, the higher layers exhibit a plateau in similarity scores, indicating that feature transformations stabilize and converge toward an optimal representation. This plateau also suggests redundancy in the higher layers, implying that removing these redundant layers could improve efficiency without significantly sacrificing performance.[4] On the other hand, for models trained on large datasets, Figure 2(d) shows a consistent ridge pattern, characterized by a rapid decay in similarity between adjacent layers. This indicates a dynamic and continuous refinement process throughout the network. Nevertheless, for both Figure 2(c, d), we observe almost all nonnegative average cosine similarity between different layers, albeit the features are almost orthogonal when the layers are far apart. Moreover, in both Figure 2(c, d), we observe a progressive increase in representation similarity as the two layers get closer; across each row or column, the cosine similarity increases as it approaches the diagonals. In appendix D, we observe similar phenomena on multi-modality models (CLIP).

To understand this phenomenon, we utilize the connection between residual network (ResNet) and dynamic system, viewing the residual update (1) as a discretization of a dynamic system [50, 27]. Specifically, Gai and Zhang [27] proved that ResNet trained with weight decay attempts to learn the geodesic curve in the Wasserstein space. Since transformer is a ResNet and weight decay is commonly applied in real-world training—e.g., DeiT [24] models are trained using the AdamW optimizer with a weight decay of 0.05—we use the following geodesic curve assumption [51].

Assumption 1 *(Geodesic curve assumption) At the terminal phase of training, the transformer with weight decay has learned the geodesic curve in Wasserstein space $\mathcal{P}(\mathbb{R}^d)$, which is induced by the optimal transport map.*

Based on this assumption, the following result shows a monotonic increase in representation similarity as the layers get closer. Proofs are given in the Appendix A.

[4] We notice concurrent work [47–49] that exploits representation similarity across layers for pruning redundant layers. For instance, Figure 2(c, d) show hidden layers obey large similarity, indicating that some of the layers can be skipped [49].

Theorem 2.1 *(Representation similarity increases monotonically across layers) Under Assumption 1, for any layers $\ell_1 < \ell_2 < \ell_3$, we have* $\mathrm{COS}(\boldsymbol{h}_{k,i}^{(\ell_1)}, \boldsymbol{h}_{k,i}^{(\ell_3)}) < \mathrm{COS}(\boldsymbol{h}_{k,i}^{(\ell_2)}, \boldsymbol{h}_{k,i}^{(\ell_3)})$.

Theorem 2.1 provides a theoretical justification for the phenomenon observed in Figure 2—for instance, the similarity to the last layer features progressively increases from shallow to deep layers. However, we note that in practice, the geodesic curve assumption may not hold precisely by a practical network, so some samples may not exhibit a strictly monotonic increase.

2.2 Analysis for Saturation Events

We now use the layer-wise representation similarity to analyze the phenomenon of *saturation events* [25]. To that goal, we first briefly introduce the neural collapse ($\mathcal{NC}$) phenomenon [10] and its connection to layer-wise representation similarity.

Neural Collapse ($\mathcal{NC}$) Roughly speaking, $\mathcal{NC}$ concerns the terminal phase of training deep networks and states that (i) within-class variable collapse ($\mathcal{NC}_1$): the last-layer features from the same class become nearly identical, i.e., $\boldsymbol{h}_{k,i}^{(L)} \to \overline{\boldsymbol{h}}_k^{(L)} = \frac{1}{n}\sum_i \boldsymbol{h}_{k,i}^{(L)}$, (ii) maximal distance ($\mathcal{NC}_2$): those from different classes become maximally linearly separable, and (iii) self-duality ($\mathcal{NC}_3$): the last-layer linear classifiers $\boldsymbol{w}_k$ align with the class-mean features $\overline{\boldsymbol{h}}_k^{(L)}$. To achieve this, deep classifiers progressively compress within-class features while enhancing the discrimination of between-class features from shallow to deep layers [15–17]. Our results show that transformer models achieve progressive compression and separation by progressively aligning the features with the last-layer classifier from shallow to deep layers.

Alignment between layer-wise cosine similarity and accuracy. Motivated by the similarity between features in shallow and deep layers, we apply the classifier to each hidden layer to obtain the layer-wise validation accuracy, which is plotted in Figure 2(a, b). We observe a high correlation between the layer-wise accuracy and the cosine similarity, with layer-wise accuracy also exhibiting a monotonic increase across layers. Our following result provides a justification for this phenomenon, demonstrating that an increase in representation similarity implies an increase in predicted probability across layers.

Theorem 2.2 *(Predicted probability increases monotonically across layers) Assume that Theorem 2.1 holds at $\boldsymbol{h}_{k,i}$, i.e.,* $\mathrm{COS}(\boldsymbol{h}_{k,i}^{(\ell+1)}, \boldsymbol{h}_{k,i}^{(L)}) > \mathrm{COS}(\boldsymbol{h}_{k,i}^{(\ell)}, \boldsymbol{h}_{k,i}^{(L)})$*, and that the last-layer features and classifers satisfy $\mathcal{NC}$. Then, the predicted probability* $[\mathit{SoftMax}(\boldsymbol{W}\boldsymbol{h}_{k,i}^{(\ell)})]_k$ *increases across layers:*

$$[\mathit{SoftMax}(\boldsymbol{W}\boldsymbol{h}_{k,i}^{(\ell+1)})]_k > [\mathit{SoftMax}(\boldsymbol{W}\boldsymbol{h}_{k,i}^{(\ell)})]_k. \tag{4}$$

Saturation events The approach of applying last-layer classification to intermediate representation is also called logit lens [52]. Recent studies [53, 54] use this method to decode hidden states into probability distributions over the vocabulary, offering mechanistic interpretability of transformers. [25] discover a phenomenon called saturation events, where the model's final predicted token becomes the top candidate in a certain shallow layer and remains unchanged across all subsequent layers. Specifically, given an input sample $\boldsymbol{x}_{k,i}$, the saturation layer $\ell_{k,i}$ for $\boldsymbol{x}_{k,i}$ is defined as the smallest layer ℓ such that

$$g(\boldsymbol{h}_{k,i}^{(1)}) \neq \ldots \neq g(\boldsymbol{h}_{k,i}^{(\ell)}) = \cdots = g(\boldsymbol{h}_{k,i}^{(L)}).$$

Results in Appendix C.1 show saturate events also happen on recently developed LLMs such as LLaMA3 [55]. Our Theorem 2.2 offers a justification for saturation events, stating that if a sample is correctly predicted at the ℓ-th layer, it will continue to be correctly predicted in subsequent layers, as the predicted probability increases progressively across layers.

To further illustrate the relation between representation similarity and saturate events, we train a DeiT-S model on both CIFAR-10 and ImageNet-1K and plot the results in Figure 3. For the same model across different datasets, we observed that higher layer-wise representation similarity COS correlates with more early saturation events, suggesting COS is a valuable metric for reflecting saturation events.